Carlos Romero Madrid

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TO: FERNANDO CUCCHIETTI & SELECTION COMMITTEE

Data Analytics & Visualization Group (DAV) · BSC-CNS

APPLICATION STATEMENT

Creative Computing & Data Storytelling · 2026

Dear Fernando and Members of the Data Analytics and Visualization Group,

Most resumes start with an enumeration of computational libraries. Let me start with a contrast: the low acoustic hum of **MareNostrum 5** solving fluid turbulence across thousands of GPU cores, against the gentle harmonic chime of an interactive kinetic sphere when my cat **Pipa** taps it across the floor.

I am a **Computational Physicist** (Double Degree in Physics & Mathematics from the University of Seville, 1st of Class MSc in Plasma Physics at Aix-Marseille) and an **HPC Scientific Software Engineer** at BSC. Every day, I profile, refactor, and scale massively parallel solvers on Europe's flagship supercomputers (MN5, Leonardo, LUMI). Yet my strongest conviction is that *if petabytes of scientific breakthroughs stay trapped in cold terminal outputs and static papers, we have only realized half of science's societal promise.*

The Domestic Data Streamers Philosophy: *Much like the visionary spatial storytelling of Domestic Data Streamers, I believe data is a living, sensory medium. Translating extreme HPC simulations into tangible, interactive public experiences turns abstract complexity into emotional human discovery.*

In my independent creative studio, I investigate how physical hardware, generative audio, and motion design transform technical data into visceral encounters. My goal is to bring this exact dual proficiency—deep mathematical and high-performance computational mastery paired with sensory empathy—directly into the **Data Analytics and Visualization Group**. Whether engineering real-time digital twins (**GDU-HPC**), communicating urban climate resilience (**AIR / REALLOCATE**), or curating interactive physical museum installations, I look forward to building experiences that captivate visitors and elevate public understanding of science.

PORTFOLIO HIGHLIGHTS · TANGIBLE COMPUTING & SCIENTIFIC OUTREACH

MNusic Engine
Sonic & Visual MN5
Real-time supercomputer telemetry translated to generative ambient audio and particle dynamics.

Pipa-Music-Ball
Bio-Kinetic Physical UX
ESP32 + 6-DoF inertial sphere translating feline kinetic vectors into microtonal acoustics.

Expo-Curator
Exhibition Staging
Open curation platform for automated wall layout, booklet design, and gallery flow.

With sincere enthusiasm and warm regards,

Carlos Romero Madrid

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